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Clinical challenges

A high-pressure situation



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Case report

An 8-year-old boy presented to an outside hospital with acute, severe visual loss in the right eye and subacute nausea, vomiting, and diplopia. The patient developed progressive nausea and vomiting 2 weeks earlier. After 1 week of nausea, he developed diplopia. The day before presentation, he experienced sudden loss of vision in the right eye. He had no significant past medical or ocular history and no family history of childhood eye disease or malignancy.

The visual acuity was hand motion OD and 20/20 OS. The pupillary response in the right eye was sluggish, with a relative afferent pupillary defect (RAPD). Extraocular motility testing was notable for a right hypertropia and an esotropia and mild abduction deficit in the right eye. On funduscopy, the right eye had 2+ disc edema without hemorrhage. Left eye examination was unremarkable.

Where is the most likely site of the causative lesion?

Comments by Dr. Michael C. Brodsky, MD: Based on the signs and symptoms at presentation, the major neurodiagnostic considerations are elevated intracranial pressure and pediatric optic neuritis. The coincident signs and symptoms of nausea, vomiting, and diplopia with an abduction deficit suggest that the intracranial pressure was elevated at the outset, and elevated intracranial pressure (with the “site” dependent on the lesion or the associated pathophysiology) can rarely produce unilateral papilledema.⁴ A history of headaches, pulsatile tinnitus, or transient visual obscurations would further support the diagnosis of elevated intracranial pressure; however, unilateral papilledema would not be associated with profound visual loss in the eye with optic disc swelling, making the diagnosis of unilateral optic neuritis much more likely.

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In this case the “site” would be the orbital optic nerve. Common causes of pediatric neuritis include neuromyelitis optica (NMO), MOGAD (MOG-associated demyelination), antecedent viral infection, multiple sclerosis, vaccinations, cat scratches, bee stings, and tick bites.⁵ The clinical history should be first directed toward eliciting evidence of these conditions. Clinical examination should include slit lamp biomicroscopy for anterior vitreous cells, which accompany many forms of papillitis.

Demyelinating optic neuritis may be accompanied by intracranial pressure elevation,²³ but this degree of elevation would probably be insufficient to produce the above-mentioned signs and symptoms; however, optic neuritis can occur in the setting of acute disseminated encephalomyelitis (ADEM), which could produce intracranial pressure elevation in conjunction with unilateral optic neuritis.⁵ The association of optic neuritis with anti-MOG antibodies is a common manifestation of pediatric optic neuritis with optic disc swelling, and this association can accompany ADEM.⁵ Similarly, neuromyelitis optica (formerly known as Devic disease) can be accompanied by a recent history of nausea, vomiting, hiccoughs, or difficulty walking.⁵

Case report (continued): At the referring hospital, the patient underwent a computed tomography (CT) scan of the head. This showed an enhancing right orbital apex mass involving the optic canal and superior orbital fissure with associated bony destruction. A complete blood count and metabolic panel were normal. High dose intravenous dexamethasone was initiated, and the patient was transferred to a tertiary-care pediatric hospital for evaluation by the ophthalmology and oncology services.

What would you do next?

Comments by Dr. Brodsky (continued): The finding of an orbital apex lesion on CT scanning changes the differential diagnosis to include an inflammatory, infectious, or neoplastic lesion involving the orbital apex.³ Inflammatory or infectious lesions would include sarcoidosis, histiocytosis, tuberculosis, or aspergillosis.³⁷ Neoplastic lesions would include leukemia, lymphoma, or rhabdomyosarcoma.^{3,21} Vasculitic disorders such as granulomatosis with polyangiitis produce orbital bony destruction and can rarely present in the pediatric population.^{33,6} The presence of bony destruction, especially when accompanied by diabetes insipidus, suggests an inflammatory lesion such as histiocytosis, which usually involves the lateral wall of the orbit,¹⁹ or sarcoidosis.^{29,24} These inflammatory diseases can cause optic disc edema intracranially, intraorbitally, or via direct infiltration of the optic disc to produce elevation.²⁶ Given the differential diagnosis, I would again perform careful slit lamp and retinal examinations to look for signs of associated uveitis. I would also perform MR imaging of the head and spine to look for additional CNS lesions (metastatic or multicentric), as well as occult neuroimaging signs of elevated intracranial pressure (flattening of the posterior sclera, distended perioptic CSF sheaths, empty sella, papillary protrusion into the vitreous, vertical tortuosity of the optic nerves, and empty sella). I would also obtain a lumbar puncture to check the opening

pressure and examine the cerebrospinal fluid for biochemical evidence of infection or leukemia.

The association of unilateral optic disc edema with an orbital lesion was recognized by Smith in 1939.³⁰ The bony lesion at the orbital apex depicted on the CT scan would be expected to cause a compressive optic neuropathy, and optic disc edema can result primarily from blockade of rapid orthograde axonal transport,¹⁶ although disc edema is generally more common and more severe when the tumor is situated more closely to the globe.³⁵ In this case, the orbital apex lesion could also involve the fourth and sixth cranial nerves to produce the ocular motility disturbances described, but the associated systemic findings of nausea and vomiting would again suggest more widespread involvement or elevated intracranial pressure.

Case report (continued): The patient underwent biopsy of his right orbital mass via an orbitotomy. The biopsy showed diagnostic features of Langerhans cell histiocytosis (LCH). This was characterized by histiocytic cells with grooved to indented nuclei and with relatively abundant eosinophilic cytoplasm (Fig. 1). The lesional histiocytic cells expressed cytoplasmic CD1a and CD207 (Langerin) immunoreactivity, confirming the diagnosis of LCH. The BRAF V600E mutation was identified in lesional tissue but was negative in peripheral blood, indicating no circulating lesional cells. The patient underwent a skeletal survey and positron emission tomography (PET) scan, that identified a single right greater sphenoid wing lesion. Magnetic resonance imaging (MRI) of the brain and orbits (Fig. 2) showed a gadolinium-enhancing mass involving the right orbital roof and planum sphenoidale and extending into the orbital apex. The mass circumferentially encased the optic nerve and involved the superior oblique, levator palpebrae superioris, and superior rectus muscles. The patient was diagnosed with isolated orbital LCH as the etiology for his orbital apex syndrome, characterized by compressive optic neuropathy and ophthalmoplegia.

The patient remained on intravenous steroids for 3 days following biopsy. A 5-day cycle of intravenous cytarabine chemotherapy was then initiated at the standard dose, 100 mg/m². This was the first cycle in a planned one-year regimen as part of a clinical trial⁸ comparing cytarabine with combined vinblastine and prednisone as frontline therapy for LCH. Two weeks later, the patient reported improved vision without diplopia. Visual acuity was 20/60 OD, and his esotropia had resolved.

The patient received identical cytarabine cycles every 3 weeks for the first 5 cycles and every 4 weeks for subsequent cycles. In follow-up neuro-ophthalmology appointments, there was gradual return of right eye vision. Following the second cytarabine cycle, the optic disc edema in the right eye had resolved. Optical coherence tomography (OCT) measurement of the retinal nerve fiber layer (RNFL) was decreased in the right eye (78 μ m) and normal in the left (91 μ m).

After 2 chemotherapy cycles, MRI showed decreased size of the orbital LCH lesion, and after 4 cycles, MRI showed near complete resolution, considered a complete response. There were no signs of recurrence or major complications until during treatment cycle 6, when examination showed a slightly elevated left optic nerve. Mean RNFL thickness was 72 μ m OD and 139 μ m OS.

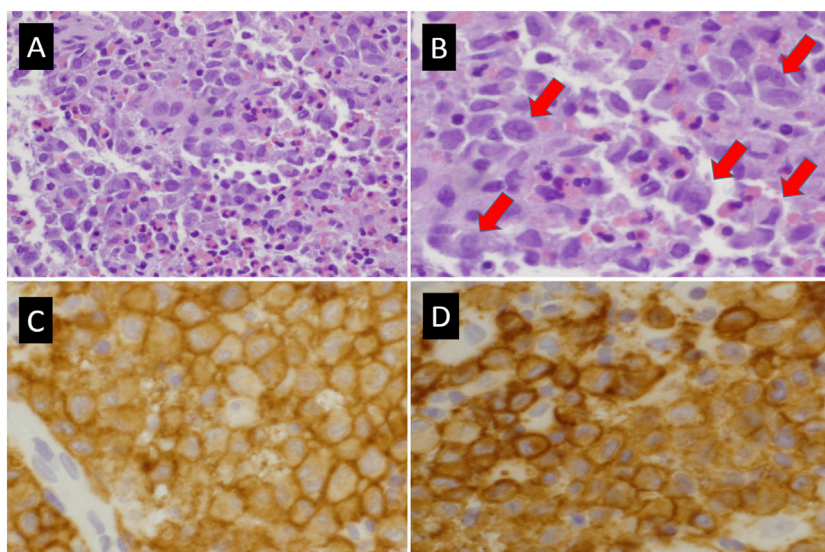


Fig. 1 – Histopathologic appearance of Langerhans cell histiocytosis. The lesion includes histiocytes with reniform to indented nuclei and relatively abundant eosinophilic cytoplasm (A and B; H&E staining). Diagnosis of LCH is confirmed with cytoplasmic immunohistochemical staining with CD1a (C; dendritic cell marker) and CD207 (D; Langerin, sensitive and specific for Langerhans cells and Langerhans cell histiocytes).

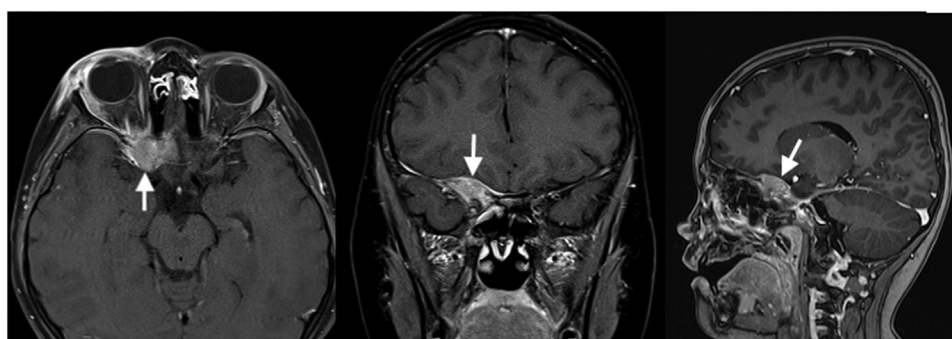


Fig. 2 – MRI (T1, postcontrast with fat suppression) revealed an enhancing mass (arrows) involving the right orbital roof, planum sphenoidale and orbital apex. The mass encased the optic nerve and involved several orbital muscles. Left: axial; middle: coronal; right: sagittal.

What are the most likely causes for the elevated retinal nerve fiber layer in the left eye?

Comments by Dr. Brodsky (continued): OCT documentation of increased RNFL thickness in the left eye in the absence of visual loss again raises the possibility of papilledema. The preservation of vision in the left eye suggests that this new optic disc swelling is not indicative of an optic neuritis, while the absence of signs and symptoms of elevated intracranial pressure with resolution in diplopia would suggest that any intracranial pressure elevation is relatively mild. Given the time course, this new disc swelling may represent the initial manifestation of papilledema from a longstanding intracranial pressure elevation (consistent with symptoms of nausea and vomiting at initial presentation), or an increase in papilledema due to disease or treatment. Signs and symptoms

of elevated intracranial pressure should again be sought. Repeat MR imaging and MR venography should be performed to rule out the (very) unlikely possibility of a new lesion causing hydrocephalus or venous sinus compression. A lumbar puncture should be performed to reassess CSF opening pressure.

Case report (continued): The patient denied visual complaints, headache, or nausea. Other findings were stable or improved from the previous examination. MRI was repeated but was unchanged from previous imaging, and the patient continued treatment; however, following treatment cycle 7, there was worsening bilateral disc edema with RNFL global values of 113 μm OD and 294 μm OS.

A lumbar puncture showed an elevated opening pressure of 51 mm Hg. Cerebrospinal fluid had mildly decreased protein with normal glucose and cell counts and negative culture and gram stain.

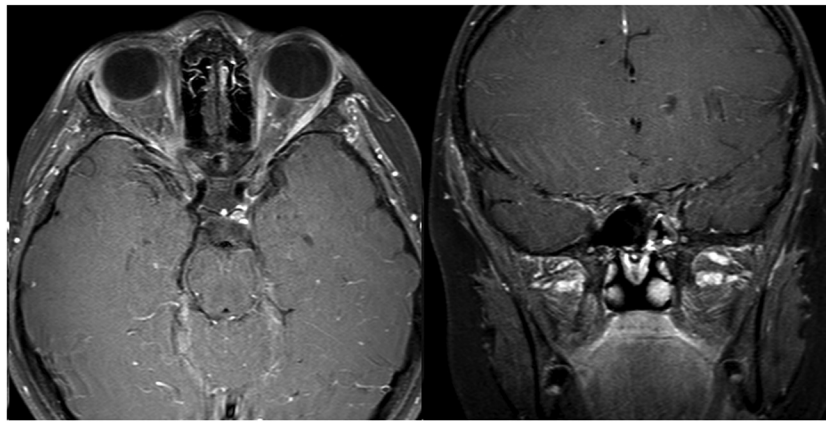


Fig. 3 – MRI (T1, postcontrast with fat-suppression) following the year-long cytarabine regimen showed complete resolution of the right orbital LCH lesion with no residual disease detected. Left: axial; right: coronal.

What is the most likely cause of the elevated intracranial pressure? What treatment would you suggest?

Comments by Dr. Brodsky (continued): If elevated intracranial pressure is found in the absence of a causative lesion, consideration should be given to the possibility that cytarabine, which is known to cause intracranial pressure elevation when administered at high doses,²⁵ may be responsible for this unusual postsript. If so, then discontinuing this drug (or reducing the dose if discontinuation is deemed inadvisable) should produce correlative improvement in the papilledema.

Case report (concluded): The elevated intracranial pressure and papilledema were attributed to cytarabine, a known cause of intracranial hypertension. The patient continued chemotherapy but was begun on acetazolamide 250 mg 4 times daily (QID), subsequently increased to 500 mg QID.

The papilledema resolved after 3 months of acetazolamide therapy. The year-long cytarabine regimen was completed without LCH recurrence or major complications. Post-treatment MRI showed no residual disease (Fig. 3). At the conclusion of therapy, visual acuity was 20/25 OD and 20/20 OS with full color vision and no RAPD or extraocular motility deficits. Acetazolamide was reduced to 500 mg twice daily, and four months later was discontinued without recurrent optic disc edema.

Discussion

Langerhans cell histiocytosis is a rare disorder characterized by proliferation of Langerhans cells histiocytes that may involve several organs systems, bone and soft tissue.¹ Although the pathophysiology of LCH is poorly understood, this proliferation is derived from myeloid precursor cells and possesses dendritic cell features. It is unclear whether LCH is best considered a neoplastic or immunodysregulatory disorder.¹ The disease occurs most commonly in children and in a spectrum from unifocal to multisystem disease.¹

Lytic bone lesions are the most common lesion in LCH,²⁷ and bone lesions of the orbit are the most common form of disease relevant to the ophthalmologist.¹⁷ Orbital involvement in LCH is fairly common, with estimates of its frequency ranging from 11% to 37.5%.^{7,9,22,34} Orbital lesions may be confined to bone or associated with a soft tissue mass and typically involve the orbital roof or lateral wall.^{17,19} Common presenting symptoms include proptosis, edema, or a visible orbital mass.^{17,19} Optic nerve involvement and vision loss with orbital LCH are rare.²²

In the current case, LCH likely arose from the sphenoid bone and involved the orbital apex, resulting in an orbital apex syndrome. Orbital apex involvement is rare in LCH, with only a handful of case reports in the literature. These include one case with an orbital apex and cavernous sinus lesion,¹³ 2 cases with orbital apex and sphenoid sinus lesions,^{28,32} and one case of an orbital apex lesion suspected to arise from the sphenoid bone.³⁶ All these patients developed optic neuropathy, with 2 developing ophthalmoplegia.^{13,36} These cases illustrate that LCH is a rare cause of a skull based lesion and may result in an orbital apex or cavernous sinus syndrome.

Isolated LCH bone lesions typically respond well to local treatment with curettage, while multisystem disease requires chemotherapy.¹⁵ Unifocal orbital or skull base lesions have traditionally been considered CNS risk lesions and require chemotherapy under oncology treatment guidelines,¹⁵ although some authors have questioned the strength of this recommendation.¹⁴ The frontline chemotherapy regimen in children is vinblastine and prednisone,¹⁵ although cytarabine has shown efficacy in other contexts¹ and is being evaluated as frontline therapy in the trial⁸ in which our patient participated.

Cytarabine (ara-C) is known to cause increased intracranial pressure. This has been reported in cases when drug administration was intravenous^{10,11,25} and intrathecal with both the standard^{2,25} and liposomal^{12,18,20,31} formulations. Given our patient's atypical demographic for idiopathic intracranial hypertension and a lack of other etiologies, we believe his condition was caused by cytarabine. The intracranial hypertension developed after prolonged exposure to cytarabine, which has been described in some of the prior reports.^{2,18,25} A high

index of suspicion for increased intracranial pressure should be maintained in such patients, as this potentially vision-threatening condition may be asymptomatic or present with nonspecific symptoms.

Cytarabine-induced intracranial hypertension may be treated effectively with acetazolamide.^{2,11,12,20,25} Our patient's excellent response to acetazolamide suggests cytarabine is not contraindicated in such scenarios and intracranial hypertension can be treated effectively even if cytarabine is continued.

Method of Literature Search

The English language literature was searched on PubMed. Search terms included "Langerhans cell histiocytosis," "orbital apex," "cytarabine," and "intracranial hypertension." Articles relevant to the current case were selected for the discussion.

Disclosures

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Conflicts of Interest

None to disclose for any author.

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ABSTRACT

Keywords:

Langerhans cell histiocytosis (LCH)
Orbital apex syndrome
Cytarabine
Intracranial hypertension

An 8-year-old boy presented with acute visual loss in the right eye and nausea, vomiting, and diplopia. Imaging revealed a right orbital apex mass. Biopsy showed Langerhans cell histiocytosis (LCH), and the patient was diagnosed with isolated orbital LCH causing an orbital apex syndrome. A 12-month cytarabine chemotherapy course was begun, during which the patient developed bilateral optic disc edema. He was diagnosed with cytarabine-induced intracranial hypertension, which was successfully treated with acetazolamide. The cytarabine course was completed with complete resolution of the LCH lesion. The ophthalmologic relevance of this rare disorder is discussed.

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